SRS — ms_06 Warehouse

1. Fungsi

Mengelola **stok gudang**, memenuhi restock request dari retail (reservasi stok lalu meminta pengiriman), dan **memesan barang ke supplier** ketika stok gudang tidak cukup, termasuk penerimaan barang (goods receipt).

2. Kebutuhan

- Stok per barang: `quantity`, `reserved`, `min_stock`; tersedia = `quantity − reserved`.
- Setiap perubahan stok tercatat di `stock_movements` (IN/OUT/RESERVE/RELEASE) — jejak audit.
- Terima `restock.requested`: bila stok cukup → reserve + `shipment.requested`; bila kurang → `WAITING_SUPPLIER` + `warehouse.stock.insufficient`.
- Pembelian ke supplier: manajer membuat purchase request → event `po.requested`.
- Goods receipt saat PO delivered menambah stok, lalu memproses ulang order yang menunggu.
- Saat `shipment.shipped`, stok dikurangi (OUT) dan reservasi dilepas.

3. Endpoint

Method	Path	Deskripsi	Peran
CRUD	`/warehouses`	Master gudang	OWNER
GET	`/stocks`	Stok gudang	WAREHOUSE_MANAGER, OWNER
POST	`/stocks/adjustments`	Penyesuaian stok manual	WAREHOUSE_MANAGER
GET	`/stock-movements`	Riwayat pergerakan	WAREHOUSE_MANAGER
GET	`/restock-orders`	Order restock dari retail	WAREHOUSE_MANAGER
POST	`/restock-orders/{id}/reject`	Tolak order	WAREHOUSE_MANAGER
POST	`/purchase-requests`	Pesan ke supplier `{supplier_id, items}`	WAREHOUSE_MANAGER
GET	`/purchase-requests`	Status pemesanan	WAREHOUSE_MANAGER
POST	`/goods-receipts`	Terima barang `{po_id, items}`	WAREHOUSE_MANAGER

4. Model Data

`warehouses`, `warehouse_stocks`, `stock_movements`, `warehouse_restock_orders`, `purchase_requests` — lihat er.puml.

5. Event

- Consume: `restock.requested`, `shipment.shipped`, `po.confirmed`, `po.shipped`, `po.delivered`.
- Publish: `shipment.requested`, `warehouse.stock.insufficient`, `po.requested`.

6. Diagram

usecase.puml · er.puml · sequence_restock_fulfillment.puml · sequence_purchase_supplier.puml · state_restock_order.puml

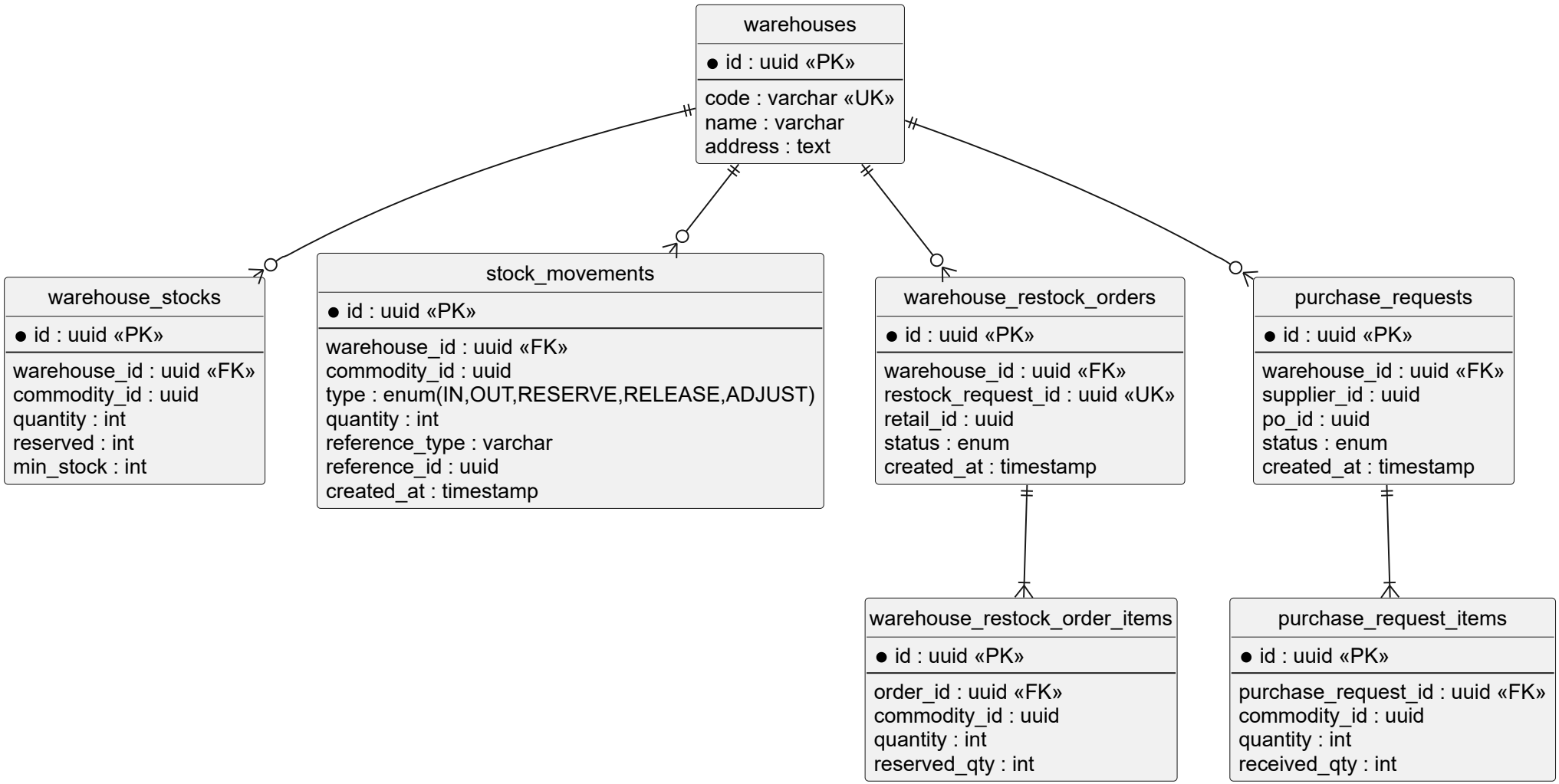
7. Cara Pembuatan

1. Database `db_warehouse`; migrasi tabel + `processed_events`, `outbox_events`; `CHECK (reserved <= quantity)`.
2. Modul stok transaksional (`FOR UPDATE`), setiap operasi menulis `stock_movements`.
3. Consumer `restock.requested` dengan logika cukup/kurang (alokasi parsial opsional: kirim sebagian, sisa menunggu supplier).
4. Modul purchase request + consumer event PO.
5. Modul goods receipt (validasi jumlah vs PO) lalu pemicu pemrosesan ulang order menunggu.
6. Uji: reservasi bersamaan, event ganda, penerimaan parsial.

Lampiran Gambar Diagram

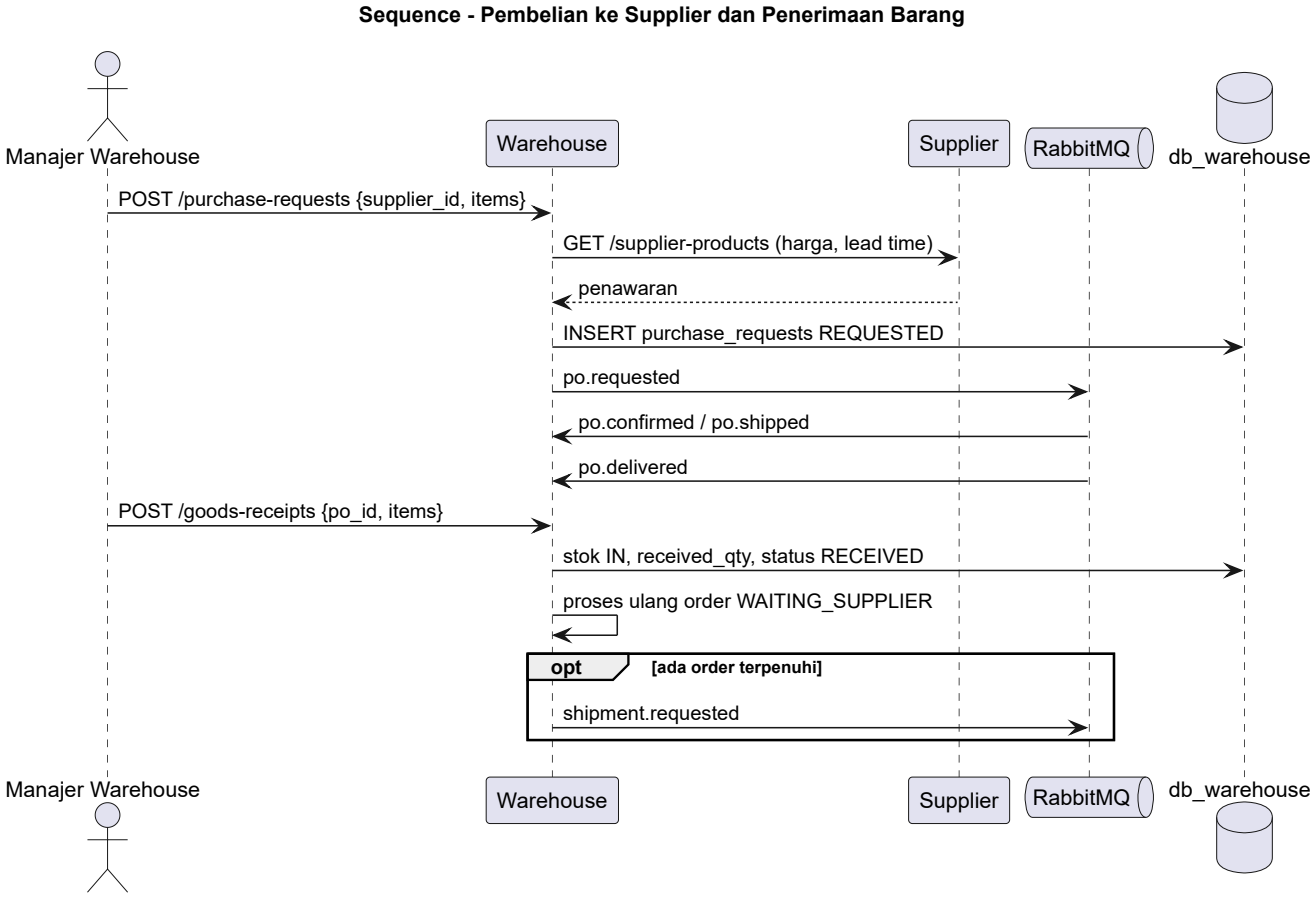
ER - Warehouse

ER - Warehouse



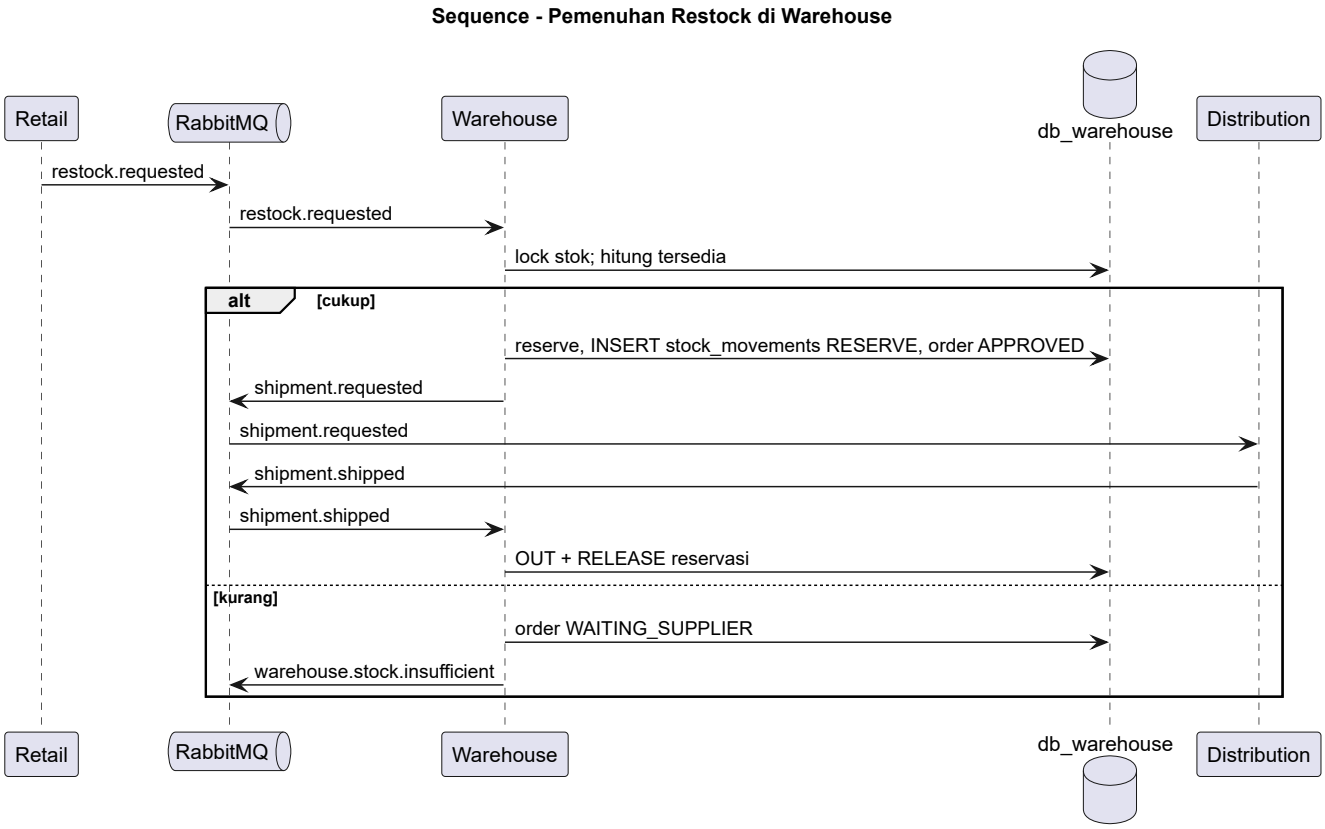
Sumber: [er.puml]

Sequence - Pembelian ke Supplier dan Penerimaan Barang



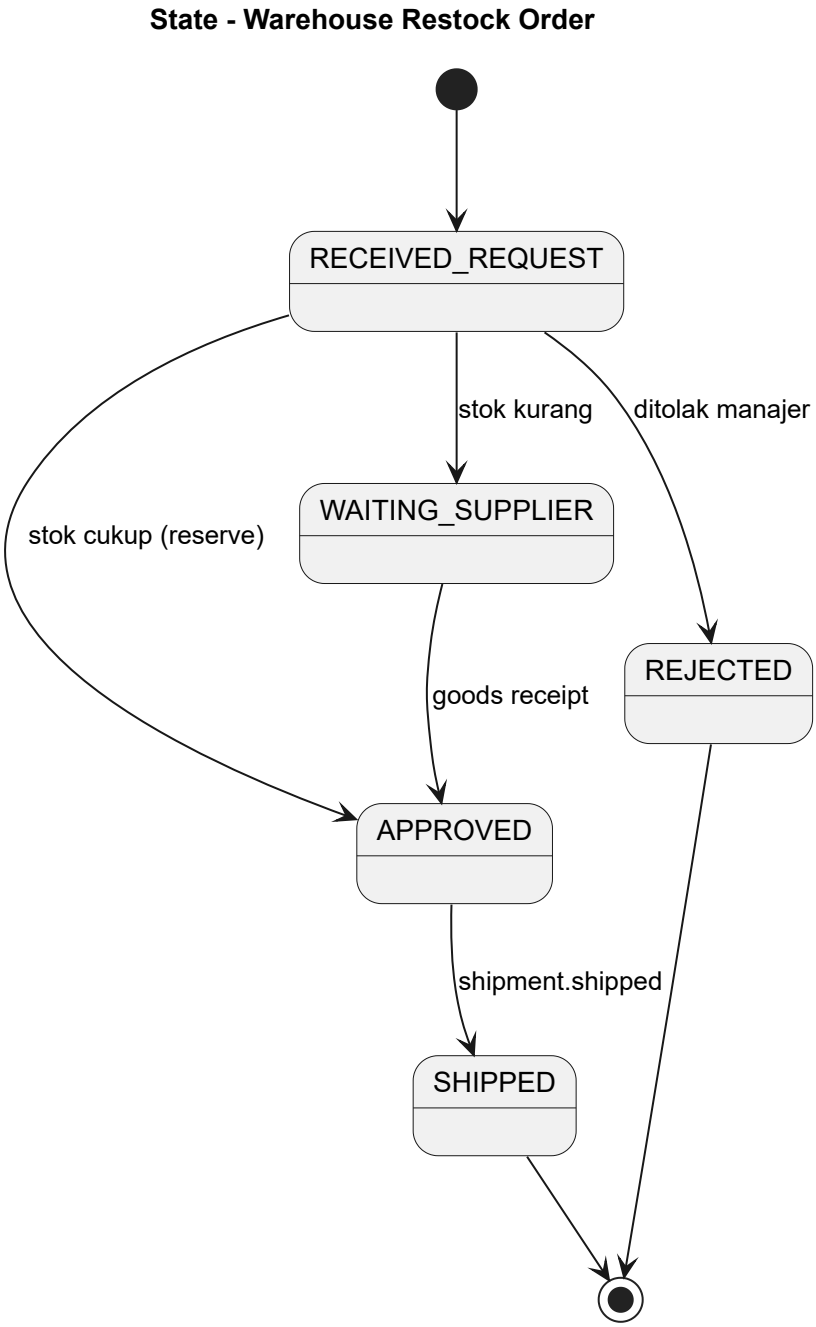
Sumber: [sequence_purchase_supplier.puml]

Sequence - Pemenuhan Restock di Warehouse



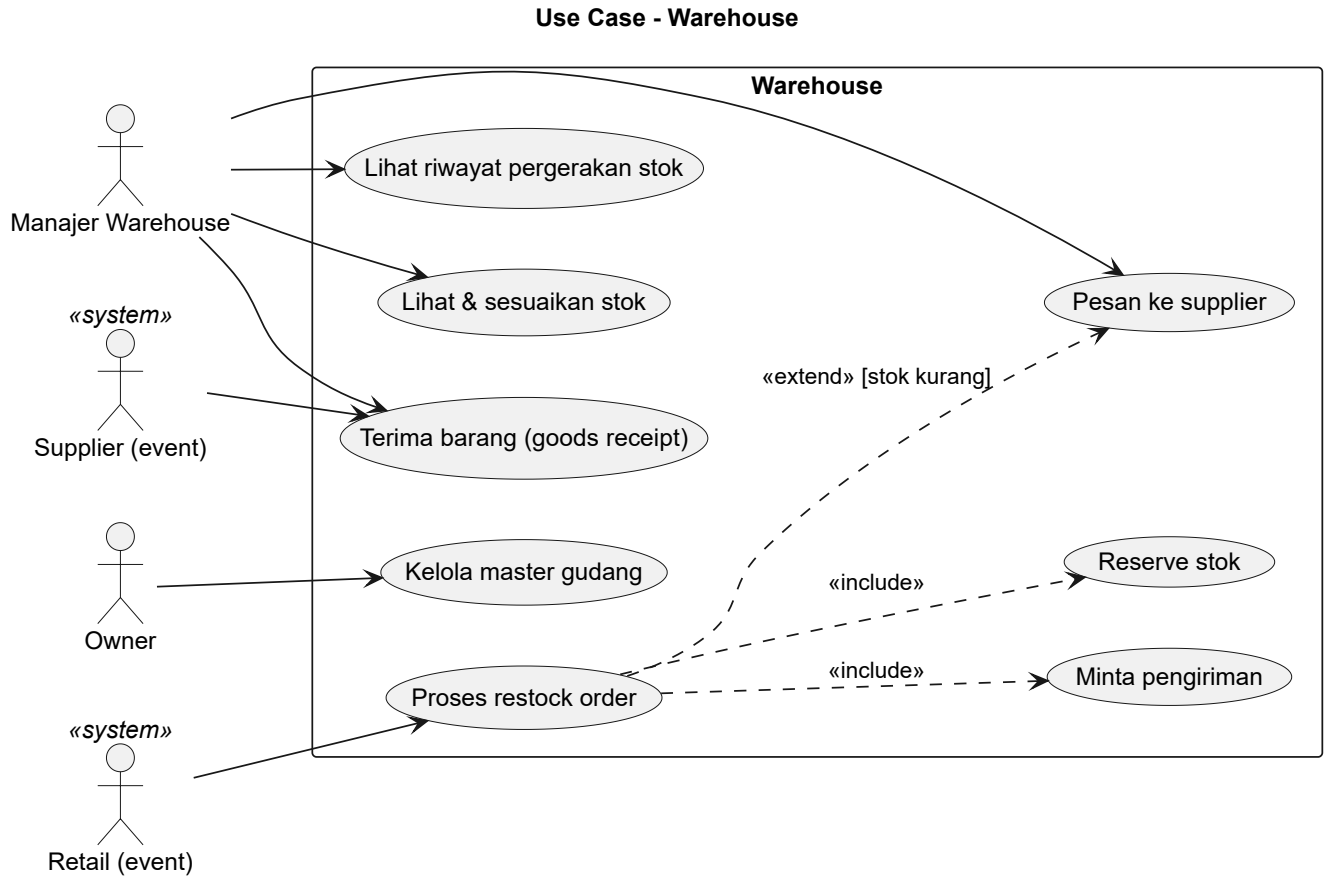
Sumber: [sequence_restock_fulfillment.puml]

State - Warehouse Restock Order



Sumber: state_restock_order.puml

Use Case - Warehouse



Sumber: usecase.puml